Venus

Earth's twin - second planet from the Sun

Overview

Venus is the second planet from the Sun and the closest planet to Earth in size and mass, which is why it is often called Earth's 'sister' or 'twin' planet. It is named after the Roman goddess of love and beauty.

Despite its similar size, Venus is extremely hostile. It is the hottest planet in the Solar System, has a crushing atmosphere and is covered by thick clouds of sulfuric acid. After the Moon, it is the brightest natural object in the night sky, and has been called the 'Morning Star' and the 'Evening Star'.

Key facts

- **Average distance from the Sun:** 0.72 AU (about 108 million km)
- **Diameter:** 12,104 km (about 95% of Earth)
- **Length of day:** 243 Earth days (rotation period, retrograde)
- **Length of year:** 225 Earth days
- **Surface gravity:** 8.87 m/s squared (about 90% of Earth)
- **Average temperature:** about 465 C
- **Number of moons:** 0
- **Axial tilt:** 177.4 degrees (rotates 'upside down')

Atmosphere and climate

The atmosphere of Venus is about 96.5% carbon dioxide, with most of the rest being nitrogen. The surface pressure is about 92 times that of Earth - similar to the pressure found about 900 m deep in Earth's oceans.

This dense carbon dioxide atmosphere creates a runaway greenhouse effect. As a result, Venus is hotter than Mercury even though Mercury is closer to the Sun. The surface temperature of about 465 C is hot enough to melt lead, and it barely changes between day and night or between the poles and the equator.

Rotation

Venus rotates very slowly and in the opposite direction to most planets (retrograde rotation), so on Venus the Sun rises in the west and sets in the east. One rotation takes 243 Earth days, which is longer than its year of 225 Earth days. Because of the combination of rotation and orbit, one solar day (sunrise to sunrise) lasts about 117 Earth days.

Surface features

Radar mapping has revealed a surface shaped by volcanism, with vast lava plains, thousands of volcanoes and relatively few impact craters, suggesting the surface is geologically young. The highest mountain range is **Maxwell Montes**, reaching about 11 km above the average surface level.

Venus has no moons and no rings.

Clouds and winds

Venus is completely covered by clouds that lie roughly 45 to 70 km above the surface. These clouds reflect most of the sunlight that hits them, which is why Venus shines so brightly in our sky, and they hide the surface from ordinary telescopes.

The upper atmosphere shows **super-rotation**: winds at cloud level race around the whole planet in about four Earth days, at speeds of around 360 km/h - far faster than the planet itself rotates. Near the surface, by contrast, winds are only a few km/h.

Interior and lost water

Venus is thought to have an internal structure similar to Earth's, with an iron core, a rocky mantle and a crust. However, it has no significant global magnetic field, probably because its very slow rotation does not drive a strong dynamo.

Scientists believe Venus may once have had liquid water, possibly even oceans. The ratio of deuterium (heavy hydrogen) to ordinary hydrogen in its atmosphere is far higher than on Earth, which suggests that large amounts of water were lost to space as the greenhouse effect ran out of control.

In 2020 researchers reported detecting phosphine gas in the clouds of Venus, a possible sign of life. Later studies questioned the result, and the detection remains disputed.

Observing Venus from Earth

Because it orbits inside Earth's orbit, Venus never appears far from the Sun and is seen only in the morning or evening sky. Through a telescope it shows phases like the Moon. Galileo observed these phases in 1610, and they were important evidence that the planets orbit the Sun rather than Earth.

Occasionally Venus passes directly between Earth and the Sun, appearing as a small black dot crossing the Sun's disk. These **transits of Venus** come in pairs eight years apart, separated by more than a century. The last two were in 2004 and 2012; the next will be in 2117 and 2125.

Exploration history

- **Mariner 2** (NASA, 1962) - first successful flyby of any planet.
- **Venera 7** (USSR, 1970) - first spacecraft to land softly on another planet and transmit data from its surface.
- **Venera 13** (USSR, 1982) - sent back the first colour images of the surface; survived about 127 minutes.
- **Magellan** (NASA, 1990-1994) - used radar to map about 98% of the surface.
- **Venus Express** (ESA, 2006-2014) - studied the atmosphere from orbit.
- **Akatsuki** (JAXA, in orbit from 2015) - Japanese orbiter studying weather and clouds.
- Planned missions: NASA's **VERITAS** and **DAVINCI**, and ESA's **EnVision**.

Did you know?

- A day on Venus is longer than a year on Venus.
- Venus is the hottest planet, even though Mercury is closer to the Sun.
- Landers on Venus survive only for minutes to about two hours before heat and pressure destroy them.